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professor

基本信息

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所属院校: 蔚山国立科技研究所

地区: South Korea

学科领域

集成电路设计与集成系统

电磁场与无线技术

电气工程及其自动化

微电子科学与工程

研究领域

电磁兼容与干扰

模拟与射频IC设计

EDA工具与电路验证

低功耗电路与能量管理

电力电子与电机驱动

微电子封装与可靠性

混合信号集成技术

电磁场理论与数值计算

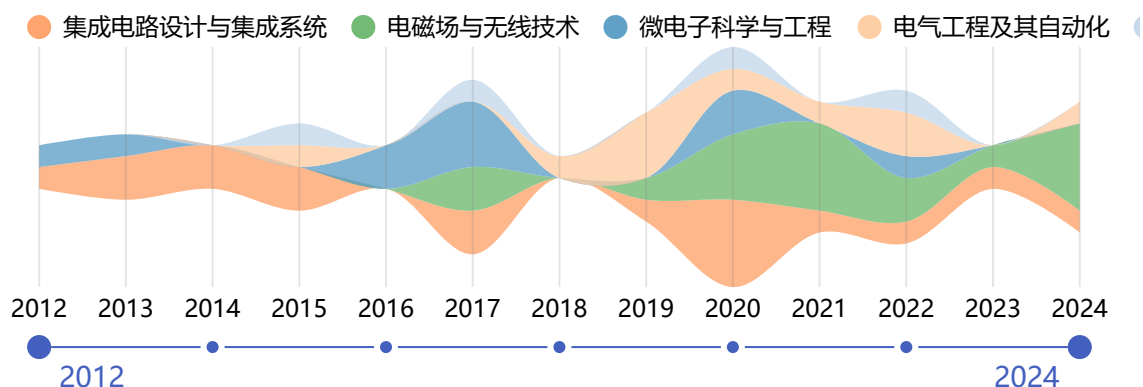
微波器件与电路

数字集成电路设计

个人简介

Jingook Kim is a professor at the Ulsan National Institute of Science and Technology. His research interests include EMCIC design, filter, ESD, and signal integrity. He has an h - index of 30 (18 since 2020) and an i10 - index of 65 (36 since 2020), with a total of 2991 citations (1272 since 2020). He has published numerous academic papers, such as "Signal integrity design for high - speed digital circuits: Progress and directions" in 2010, which has been cited 329 times.

近年发表的论文



Characterization and Application of Improved Oscilloscope IC for System Diagnosis of ESD and HPEM Effects

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Lee, Kyunghoon; Jeong, Sangyeong; Choi, Wooshin; Choi, Jung-Hwan; Kim, Jinguok

被引用次数: 0

2024

Quantitative Analysis and Prediction of EMI From Spread Spectrum of Differential Random Signaling Based on IC Source Modeling

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Bae, Byungjin; Kim, Jinguok

被引用次数: 0

2024

Prediction and Analysis of Radiated EMI From a Wafer-Level Package Based on IC Source Modeling

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Cho, Jung Hoon; Jeong, Sangyeong; Kim, Jun-Bae; Ihm, Jeong Don; Kim, Jinguok

被引用次数: 1

2024

Extraction of Noise Source Impedance of an operating high-power converter with high-EMI noises by increasing SNR using an amplifier

2024 INTERNATIONAL SYMPOSIUM AND EXHIBITION ON ELECTROMAGNETIC COMPATIBILITY, EMC EUROPE 2024

Kang, Dongwan; Jeong, Sangyeong; Kweon, Hyck Su; Kwack, Young Hwan; Kim, Jinguok

被引用次数: 0

2024

One-Chip Active EMI Filter With Integrated Buck Converter and Self-Malfunction Detection for CE Noise Reduction

IEEE TRANSACTIONS ON POWER ELECTRONICS

Jeong, Sangyeong; Kim, Jinguok

被引用次数: 1

2023

Detection Circuits for System-Level Signal and Power Noises Due to High-Power Electromagnetic Pulses

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Paek, Joon Ki; Lee, Kyunghoon; Kim, Jinguok

被引用次数: 2

2022

A Proposed On-Die Oscilloscope for Monitoring of Power Noise Waveform Inside IC Due to Transient Stress Events

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Uulu, Zakirbek Mamatair; Cho, Jung-Hoon; Bae, Bumhee; Kim, Jingook

被引用次数: 1

2022

Efficient Computation of Electromagnetic Coupling From Various Types of Antennas Based on Reciprocity Theorem

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Kwak, Kyungjin; Moon, Hwansoo; Kim, Cheolhan; Kim, Jingook

被引用次数: 2

2022

Automatic SPICE-Integrated Reinforcement Learning for Decap Optimization for EMI and Power Integrity

2022 IEEE INTERNATIONAL SYMPOSIUM ON ELECTROMAGNETIC COMPATIBILITY & SIGNAL/POWER INTEGRITY, EMCSI

Kim, Jingook; Jeong, Sangyeong; Kim, Jun-Bae; Ihm, Jeong Don

被引用次数: 2

2022

A Low EMI characteristic of LPDDR5 SDRAM with Edgeplaced PADs and Short Re-Distribution Lines

2022 IEEE INTERNATIONAL SYMPOSIUM ON ELECTROMAGNETIC COMPATIBILITY & SIGNAL/POWER INTEGRITY, EMCSI

Kim, Jun-Bae; Kwon, Chang Ki; Park, Sangwook; Sung, Yoo-Chang; Ihm, Jeong Don; Kim, Wooil; Jin, Jungho; Hwang, Sangjoon; Yoo, Changsik; Choi, Jung-Hwan; Kim, Jingook

被引用次数: 1

2022

Accurate and Efficient Computation of System-Level ESD Noise Waveforms in ISO 10605 Standard Using Decomposition Method and Split-Domain Approach

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Bae, Byungjin; Kwak, Kyungjin; Hwang, Joosung; Kim, Euihyuk; Kim, Jingook

被引用次数: 3

2022

Frequency-Dependent Bearing Voltage Model for Squirrel-Cage Induction Motors

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Yea, Manje; Kim, Jingook; Han, Ki Jin

被引用次数: 4

2022

Proposed ESD Models of Dust Cleaners and Analysis of ESD Failures in Automotive Headlamps

2021 43RD ANNUAL EOS/ESD SYMPOSIUM (EOS/ESD)

Bae, Byungjin; Hwang, Jeong; Kim, Cheol-Han; Moon, Hwan-Soo; Kim, Jingook

被引用次数: 0

2021

Optimization of PDN decoupling capacitors for EMI Reduction based on Deep Reinforcement Learning

2021 JOINT IEEE INTERNATIONAL SYMPOSIUM ON ELECTROMAGNETIC COMPATIBILITY, SIGNAL & POWER INTEGRITY, AND EMC EUROPE (EMC+SIPI AND EMC EUROPE)

Lee, Changjong; Jeong, Sangyeong; Kim, Jingook; Kim, Jun-Bae; Ihm, Jeong Don

被引用次数: 1

2021

Field source extraction of an ESD generator and its application to system-level ESD analysis in a solid-state storage system

JOURNAL OF ELECTROMAGNETIC WAVES AND APPLICATIONS

Park, Junsik; Lee, Wooryong; Kim, Namsu; Kang, Bonggyu; Ryu, Chunghyun; Lee, Jongsung; Jo, Cheolgu; Kim, Jingook

被引用次数: 1

2021

Stator Impedance Modeling Platform for the Electromagnetic Compatibility Aware Design of 3.7-to 7.5-KW Squirrel-Cage Induction Motors

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Ryu, Younggon; Yea, Manje; Kim, Jingook; Han, Ki Jin

被引用次数: 11

2021

A Customized Integrated Circuit for Active EMI Filter With High Reliability and Scalability

IEEE TRANSACTIONS ON POWER ELECTRONICS

Jeong, Sangyeong; Park, Junsik; Kim, Jingook

被引用次数: 12

2021

A Low-Voltage Microwave Plasma Ionizer With Reduced Electric Overstress Due to Displacement Current Coupling From a High-AC Voltage Source

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Bae, Byungjin; Kim, Jingook

被引用次数: 1

2021

Accuracy investigation of equivalent dipole arrays for near-field estimation in presence of shielding or dielectric structures

MICROWAVE AND OPTICAL TECHNOLOGY LETTERS

Kwak, Kyungjin; Bae, Tae-il; Hong, Kichul; Kim, Hyungsoo; Kim, Jingook

被引用次数: 5

2020

Numerical Verification of Dielectric Contactor as Auxiliary Loads for Measuring the Multi-Port Network Parameter of Vertical Interconnection Array

IEEE ACCESS

Bae, Byungjin; Kim, Jingook; Han, Ki Jin

被引用次数: 1

2020

A Cost-Effective Structure for Secondary Discharge Control to Improve System-Level ESD Immunity of a Mobile Product

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Park, Junsik; Lee, Jongsung; Jo, Cheolgu; Seol, Byongsu; Kim, Jingook

被引用次数: 3

2020

A Balanced Feedforward Current-Sense Current-Compensation Active EMI Filter for Common-Mode Noise Reduction

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Shin, Dongil; Jeong, Sangyeong; Baek, Youngjin; Park, Chansoo; Park, Gwigeun; Kim, Jingook

被引用次数: 16

2020

Measurement-Based Validation of Integrated Circuit Transient Electromagnetic Event Sensors

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Marathe, Shubhankar; Patnaik, Abhishek; Mi, Rui; Ghosh, Kaustav; Kim, Jingook; Pommerenke, David; Beetner, Daryl G.

被引用次数: 2

2020

Improved Target Impedance Concept With Jitter Specification

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Sun, Yin; Kim, Jingook; Ouyang, Muqi; Hwang, Chulsoon

被引用次数: 9

2020

Measurement and Analysis of System-Level ESD-Induced Jitter in a Delay-Locked Loop

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Jeong, Myeongjo; Shin, Minchul; Kim, Jinwoo; Seung, Manho; Lee, Seokkiu; Kim, Jingook

被引用次数: 7

2020

Spread Spectrum Technique for Decreasing EM Noise in High-Frequency APWM HB Resonant Converter With Reduced EMI Filter Size

IEEE TRANSACTIONS ON POWER ELECTRONICS

Park, Hwa-Pyeong; Jeong, Sangyeong; Kim, Mina; Kim, Jingook; Jung, Jee-Hoon

被引用次数: 16

2019

A Transformer-Isolated Common-Mode Active EMI Filter Without Additional Components on Power Lines

IEEE TRANSACTIONS ON POWER ELECTRONICS

Jeong, Sangyeong; Shin, Dongil; Kim, Jingook

被引用次数: 24

2019

Measurement and Analysis of Statistical IC Operation Errors in a Memory Module Due to System-Level ESD Noise

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Park, Myungjoon; Park, Junsik; Choi, Joungeul; Kim, Jinwoo; Jeong, Seonghoon; Seung, Manho; Lee, Seokkiu; Kim, Jingook

被引用次数: 15

2019

A Proposed Terminal-Ground EMI Filter for Reduction of Conducted Emissions Considering Cable Radiation and Safety

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Jeong, Sangyeong; Kwak, Kyungjin; Park, Gwigeun; Baek, Youngjin; Park, Dongmin; Kim, Jingook

被引用次数: 2

2019

Accuracy Investigation of a Neuromorphic Machine Learning System Due to Electromagnetic Noises Using PEEC Model

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Lee, Wooryong; Kim, Jingook

被引用次数: 13

2019

Precise Analytical Model of Power Supply Induced Jitter Transfer Function at Inverter Chains

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Kim, Heegon; Kim, Jingook; Fan, Jun; Hwang, Chulsoon

被引用次数: 20

2018

Quantified Design Guidelines of a Compact Transformerless Active EMI Filter for Performance, Stability, and High Voltage Immunity

IEEE TRANSACTIONS ON POWER ELECTRONICS

Shin, Dongil; Jeong, Sangyeong; Kim, Jingook

被引用次数: 38

2018

Statistical Analysis for Pattern-Dependent Simultaneous Switching Outputs (SSO) of Parallel Single-Ended Buffers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Jingook

被引用次数: 27

2017

IC Failure Analysis Due to Charged Board Events by Measurements and Modeling of Discharging Currents Through IC Pins

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Park, Junsik; Lee, Jongsung; Jo, Cheolgu; Seol, Byongsu; Kim, Jingook

被引用次数: 7

2017

Sensitivity Analysis of a Circuit Model for Power Distribution Network in a Multilayered Printed Circuit Board

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Shringarpure, Ketan; Pan, Siming; Kim, Jingook; Fan, Jun; Achkir, Brice; Archambeault, Bruce; Drewniak, James L.

被引用次数: 17

2017

Fast and Accurate Calculation of System-Level ESD Noise Coupling to a Signal Trace by PEEC Model Decomposition

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Park, Junsik; Lee, Jongsung; Seol, Byongsu; Kim, Jingook

被引用次数: 10

2017

Quantified Design Guides for the Reduction of Radiated Emissions in Package-Level Power Distribution Networks

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Shin, Dongil; Kim, Namsu; Lee, Jongjoo; Park, Youngwoo; Kim, Jingook

被引用次数: 12

2017

High-Frequency Testing of Vertical Interconnection Array Using Indirect Contact Probing Method With an Improved Calibration

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Jeong, Jongwoo; Kim, Jingook; Kang, No-Weon; Han, Ki Jin

被引用次数: 1

2016

Formulation and Network Model Reduction for Analysis of the Power Distribution Network in a Production-Level Multilayered Printed Circuit Board

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Shringarpure, Ketan; Pan, Siming; Kim, Jingook; Fan, Jun; Achkir, Brice; Archambeault, Bruce; Drewniak, James L.

被引用次数: 31

2016

Indirect Contact Probing Method for Characterizing Vertical Interconnections in Electronic Packaging

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Jeong, Jongwoo; Kim, Jingook; Kang, No-Weon; Han, Ki Jin

被引用次数: 2

2015

An Enhanced Statistical Analysis Method for I/O Links Considering Supply Voltage Fluctuations and Intersymbol Interference

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Kim, Jingook; Lee, Jongjoo; Park, Eunhyeong; Park, Youngwoo

被引用次数: 13

2015

Design Criteria and Error Sensitivity of Time-Domain Channel Characterization (TCC) for Asymmetry Fixture De-Embedding

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Yoon, Changwook; Tsiklauri, Mikheil; Zvonkin, Mikhail; Chen, Qinghua Bill; Razmadze, Alexander; Aflaki, Aman; Kim, Jingook; Fan, Jun; Drewniak, James L.

被引用次数: 12

2015

Analytical Calculation of Jitter Probability Density at Multistage Output Buffers Due to Supply Voltage Fluctuations

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Park, Eunhyeong; Kim, Hyungsoo; Shim, Jongjoo; Kim, Yong-Ju; Kim, Yun-Saeng; Kim, Jingook

被引用次数: 8

2015

Analysis and Design Guide of Active EMI Filter in a Compact Package for Reduction of Common-Mode Conducted Emissions

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Shin, Dongil; Kim, Sungnam; Jeong, Geunseok; Park, Jaesu; Park, Joungwook; Han, Ki Jin; Kim, Jingook

被引用次数: 63

2015

Analytical Probability Density Calculation for Step Pulse Response of a Single-Ended Buffer With Arbitrary Power-Supply Voltage Fluctuations

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Jingook; Lee, Junho; Cho, Sunki; Hwang, Chulsoon; Yoon, Changwook; Fan, Jun

被引用次数: 29

2014

Closed-Form Expressions for the Noise Voltage Caused by a Burst Train of IC Switching Currents on a Power Distribution Network

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Kim, Jingook; Lee, Jongjoo; Ahn, Seungyoung; Fan, Jun

被引用次数: 19

2014

Capacitance Calculation for Via Structures Using an Integral Equation Method Based on Partial Capacitance

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Wang, Hanfeng; Pan, Siming; Kim, Jingook; Ruehli, Albert E.; Fan, Jun

被引用次数: 5

2013

Improved Target Impedance for Power Distribution Network Design With Power Traces Based on Rigorous Transient Analysis in a Handheld Device

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Kim, Jingook; Takita, Yuzo; Araki, Kenji; Fan, Jun

被引用次数: 28

2013

Analytical Transfer Functions Relating Power and Ground Voltage Fluctuations to Jitter at a Single-Ended Full-Swing Buffer

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Hwang, Chulsoon; Kim, Jingook; Achkir, Brice; Fan, Jun

被引用次数: 39

2013

Closed-Form Expressions for the Maximum Transient Noise Voltage Caused by an IC Switching Current on a Power Distribution Network

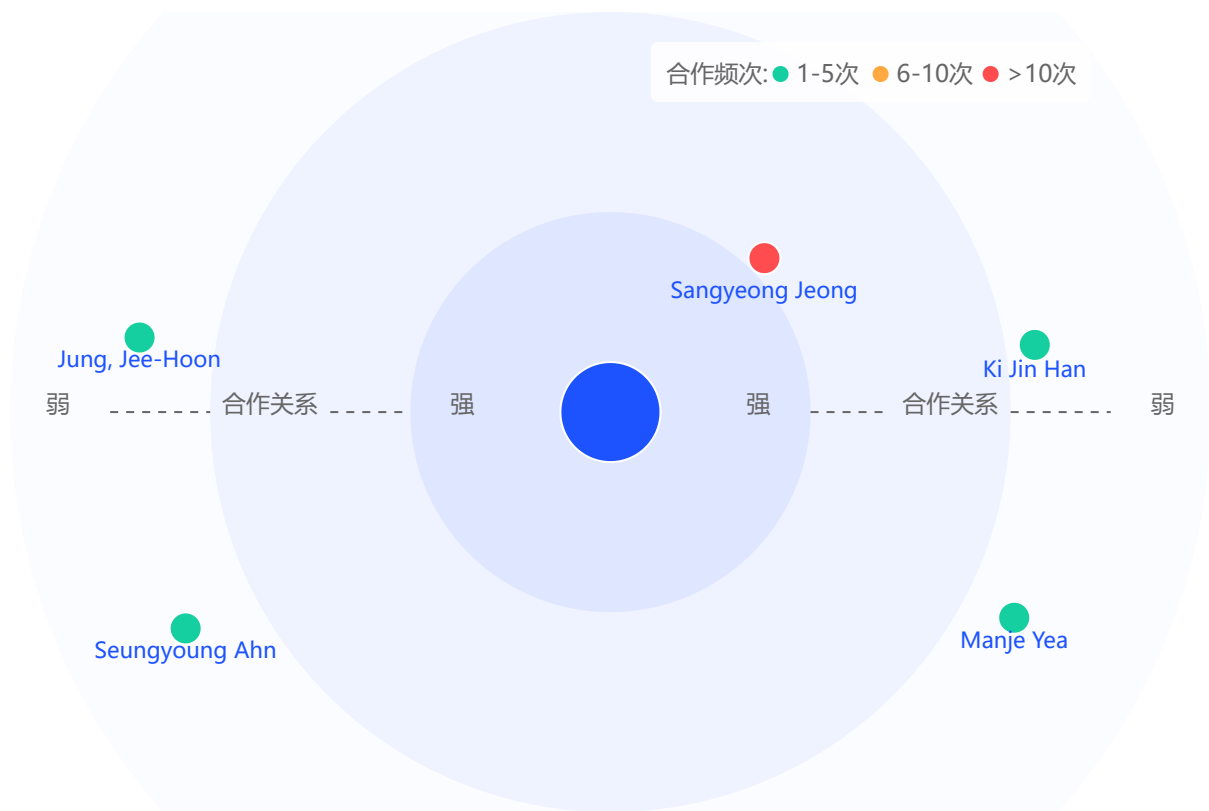
IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Kim, Jingook; Li, Liang; Wu, Songping; Wang, Hanfeng; Takita, Yuzo; Takeuchi, Hayato; Araki, Kenji; Fan, Jun; Drewniak, James L.

被引用次数: 39

2012

学术合作



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